

# Text Handoff Lauanders Uncertainty; A Mapped KV Cache Keeps More of It

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## Abstract

When one language model hands work to another through text, the receiver gets the sender’s answer but not how sure the sender was: a text handoff is a point estimate, and the receiver treats it as a fact. We ask whether handing over the sender’s key–value (KV) cache instead keeps that uncertainty. Between two frozen models of different size (K2-Horizon-3.7B  $\rightarrow$  7B) we train a per-head map from the small model’s cache to the large model’s and test on SQuAD passages that are clean, contradicted, or have their answer removed. Through the mapped cache, the receiver’s confidence tracks the *sender’s* semantic entropy nearly as well as when the receiver reads the text itself (AUROC .59–.62 vs .60–.67) and better than a text handoff carrying the sender’s answer and a confidence word (.54–.59); when the answer is removed, the cache path loses 54% of the confidence the text path loses, the verbal handoff 30%. The precondition, that the cache carries content at all, is established with controls: a closed-form map reaches 22% on a binding test (chance 12.5%), a trained gated residual 63–73%; the receiver follows a cache from the wrong episode 64% of the time; a projector trained with a deranged sender answers at 23%; on SQuAD the mapped cache reaches 53.0 F1 against 54.7 for reading the passage. The recipe crosses model families when KV geometry matches (Qwen3-4B  $\rightarrow$  7B: 71.7% mean over three seeds) and fails when it does not (K2-0.9B: 17%). Two receiver-side training objectives, a KL to the receiver’s text path and matching the sender’s entropy, did not recover the sender’s spread; we say why. We release the maps and the tests.

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## 1. Introduction

Edge–cloud agent stacks run a small model on device and escalate to a large one when the small model is unsure. Escalation today means writing a message: the small model’s answer, perhaps with a confidence word, and the large model reads it and continues. The message is a point estimate. Two senders with the same best guess and very different spread send the same words, so whatever the small model knew about its own uncertainty is discarded at the boundary, and the receiver treats a confident-looking guess as a fact—the “confidence laundering” that Shi et al. (2026) describe. If the large model could instead continue from the small model’s internal state, the handoff would carry what the small model had computed, including its spread. This paper measures how much of the sender’s uncertainty survives each kind of handoff, and finds that a mapped KV cache keeps more of it than text with a confidence word.

Before uncertainty can cross, content must. Given two frozen models of different size that share a tokenizer and KV geometry (K2-Horizon-3.7B and 7B: 36 layers, 8 KV heads, head dimension 128), can a learned map from the small model’s cache to the large model’s cache carry a *specific fact* that only the small model saw? We call this binding transfer, and we test it with a control for the standard objection that “latent gains” are extra compute rather than communication. The content results are the foundation; the uncertainty result is what the foundation is for.

**Contributions.** (1) A sender-uncertainty test on SQuAD with clean, contradicted and answer-removed passages, and the finding that a mapped-cache handoff tracks the sender’s semantic entropy better than a text handoff with a confidence word and loses less confidence when the answer is gone (Section 4), with a short argument for what a text handoff discards (the theory box). (2) A binding-transfer test (Noma) with six arms that isolates content from shape and compute, and a two-stage projector—closed-form ridge over correlation-selected source layers plus a gated residual trained on the receiver’s continuation loss—that reaches 63–73% binding transfer where the ridge alone reaches 22% (Section 3). (3) Controls that bound what the receiver actually gets from the cache: a deranged-episode cache, a deranged-sender projector, layer-band injection, and the

Table 1. Binding transfer, 8 names  $\times$  8 values,  $n = 1000$  episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

| projector                 | none | text | raw | project     | derange | follow      | content |
|---------------------------|------|------|-----|-------------|---------|-------------|---------|
| ridge top-3 (closed form) | 10.6 | 100  | 14  | 21.7        | 11.4    | 18.2        | 10.3    |
| + residual, 1500 steps    | 10.6 | 100  | 14  | <b>63.0</b> | 4.6     | <b>63.7</b> | 58.4    |
| residual, deranged sender | 10.6 | 100  | 14  | 23.4        | 11.3    | 22.7        | 12.1    |

demonstration that continuation retention, the usual headline, cannot tell a channel from a prompt (Section 5). (4) Cross-family, cross-context and message-cost results that say where the channel works and what it costs.

## 2. Setup

**Models.** Sender K2-Horizon-3.7B, receiver K2-Horizon-7B, both frozen, bf16, RoPE  $\theta = 10^7$ . We capture the sender’s pre-RoPE K and raw V at every layer, map them, apply RoPE at the receiver, and insert the result as the receiver’s cache for the prefix positions. The receiver then reads only the question.

**Projector.** Stage 1: for each receiver layer  $\ell$  and KV head  $h$ , a ridge regression from the concatenated  $[K; V]$  of the sender (256-d per head) to the receiver’s  $[K; V]$ ; source layers are the three sender layers whose K/V correlate best with receiver layer  $\ell$  (“top-3”), which beat the aligned-layer map in a first pass. Stage 2: a gated SwiGLU residual (bottleneck 128, 151 M parameters) on top of the ridge output, trained with AdamW for 1,500 steps, effective batch 16 sequences of 1,024 FineWeb-Edu tokens, prefix 512, on the receiver’s cross-entropy over the continuation. Nothing in training mentions the tests.

**Noma binding test.** Each episode assigns one of eight single-token values to eight nonce names (“Noma is amber. Vela is indigo. ...”). Only the sender reads the facts. The receiver is asked “What colour is Vela?” and we score the eight values. Arms: *none* (question only), *text* (receiver reads the facts itself), *self* (receiver’s own cache re-injected; equals *text* to  $10^{-4}$ ), *raw* (sender cache injected without a map), *project* (mapped cache), *derange* (mapped cache from another episode with the same layout but a different value for the queried name). Content transfer = project – derange. For *derange* we also report the follow rate: how often the answer is the *other* episode’s value.

**Passages.** SQuAD v1.1 validation: the sender reads the passage ( $\leq 512$  tokens), the receiver reads only the question and answers greedily ( $\leq 8$  tokens, stopped at newline); token-F1 against the gold spans. The uncertainty protocol built on it is given in Section 4.

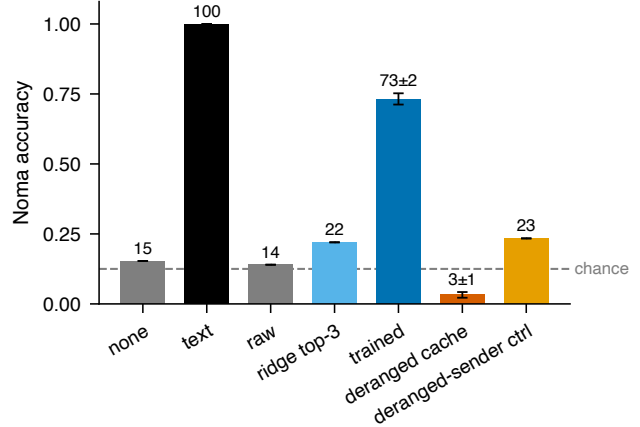


Figure 1. Noma accuracy by arm on the 6-name variant ( $n = 300$ ). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ( $n = 1000$ , 8-name variant). Dashed line: chance (12.5%).

Table 2. Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere.  $n = 500$ , 6-name variant; accuracy in percent.

| layers   | 0–5 | 0–11 | 0–17 | 12–23 | 24–35 | 30–35 | 18–35 | all  |
|----------|-----|------|------|-------|-------|-------|-------|------|
| accuracy | 7.2 | 8.0  | 10.4 | 28.6  | 30.8  | 7.6   | 51.8  | 63.0 |

## 3. Content Transfer: The Precondition

**Binding transfer.** Table 1 and Figure 1 give the main content result. On the original 6-name variant ( $n = 300$ ) three training seeds give 71.3 / 73.0 / 75.3% (mean 73.2, sd 2.0). Two ablations on the same variant (Appendix A): freezing the ridge and training only the residual gives 59.0%, so letting the linear map move during training is worth  $\sim 14$  points; replacing the correlation-selected top-3 source layers with the aligned-layer map gives 32.7%, so which sender layers feed each receiver layer is the single largest design choice. Matching shape alone (*raw*) does nothing. The follow rate is the decisive number: when the receiver is handed the wrong episode’s cache it answers with the wrong episode’s value 64% of the time, and its accuracy falls below chance. The receiver believes the cache. With the full 8-name pool and 500 episodes, the seed-2 projector gives 65.8% and 67.6% on two episode seeds (ridge: 20.4%, 20.6%). On 256 fresh held-out sequences deduplicated against training text, continuation retention is 1.19 for the trained projector and  $-0.38$  for the ridge. The same protocol with cities and animals gives 61.8% and 67.2% (Appendix B), and padding the facts with up to 512 filler tokens does not reduce accuracy (61.0 / 57.0 / 62.4 / 68.8% at 64 / 128 / 256 / 512; Appendix C).

**Where the binding lives.** Table 2 (plotted in Appendix C,

Table 3. Real passages: SQuAD v1.1 validation,  $n = 300$  (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

| receiver sees         | none  | text  | raw   | ridge | trained     | deranged-sender ctrl | trained, derange arm |
|-----------------------|-------|-------|-------|-------|-------------|----------------------|----------------------|
| F1                    | 16.6  | 54.7  | 0.6   | 14.3  | 53.0 (51.8) | 15.1                 | 13.8                 |
| gold log-prob / token | -3.57 | -0.90 | -11.2 | -3.88 | -1.74       | -3.77                | -4.18                |

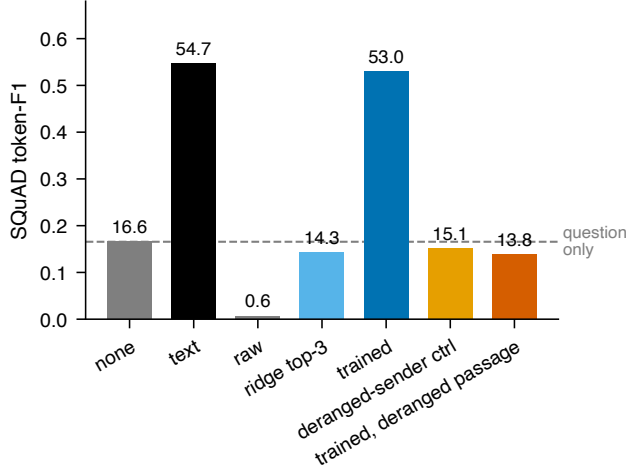


Figure 2. SQuAD token-F1 by arm ( $n = 300$ ). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage’s cache all sit at or below question-only.

Figure 8b): the first half of the network carries nothing usable on its own; the second half alone carries most of it, and no six-layer band suffices. The binding is read out from a distributed set of later layers.

**Real passages.** Table 3 and Figure 2 apply the same projectors, never trained on questions, to SQuAD. On F1 the trained projector recovers 96% of the gap between question-only and reading the passage (53.0 at  $n = 300$ ; 51.8 at  $n = 600$ ); on gold log-probability it recovers 69%. The wrong passage’s cache (derange), the ridge map, and the deranged-sender projector all sit at or below question-only. The channel carries the passage, not a bias toward answering.

**Across pairs and families.** Table 4 and Figure 3. The channel is not one-way: with the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate—the same pattern at slightly lower accuracy. With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128—matched KV geometry—but a different vocabulary and pre-training), receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge 18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7 / 70.3 / 68.0% over three train-

Table 4. Four sender–receiver pairs, same recipe. Noma 6-name variant,  $n = 300$ , chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ( $n = 300$ ; text ceiling 54.7). K2-Horizon unless named. Qwen3-4B row: proj. and F1 are mean  $\pm$  sd over three training seeds (76.7 / 70.3 / 68.0; F1 48.0 / 51.9 / 44.9); der. is the worst seed, follow is seed 0. The raw arm is undefined for K2-0.9B (head dimension 64).

| sender $\rightarrow$ receiver | ridge top-3 |       |      |        | + trained residual |            |        | F1             |
|-------------------------------|-------------|-------|------|--------|--------------------|------------|--------|----------------|
|                               | raw         | proj. | der. | follow | proj.              | der.       | follow |                |
| 3.7B $\rightarrow$ 7B         | 14.0        | 22.0  | 10.7 | 30.3   | 71.3               | 4.3        | 71.7   | 53.0           |
| 7B $\rightarrow$ 3.7B         | 11.7        | 29.0  | 12.3 | 31.3   | 63.3               | 5.7        | 63.0   | —              |
| Qwen3-4B $\rightarrow$ 7B     | 15.7        | 18.0  | 13.0 | 17.0   | 71.7 $\pm$ 4.5     | $\leq$ 4.7 | 80.3   | 48.3 $\pm$ 3.5 |
| 0.9B $\rightarrow$ 7B         | n/a         | 13.3  | 13.0 | 10.7   | 17.0               | 14.0       | 11.3   | 24.2           |

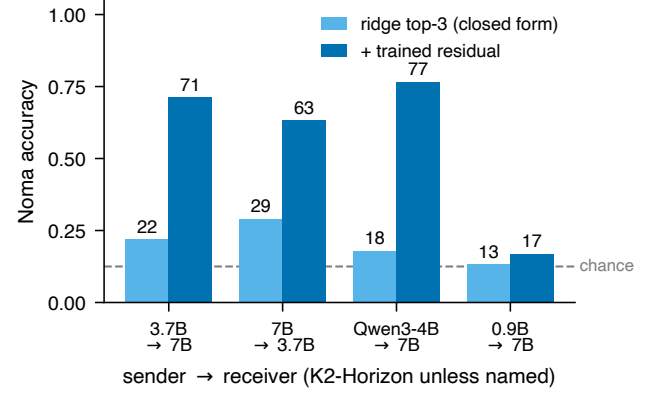


Figure 3. Noma accuracy (*project* arm) for the closed-form ridge map and the trained projector on four sender–receiver pairs ( $n = 300$ , 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

ing seeds (mean 71.7, sd 4.5; deranged  $\leq$  4.7%, follow rate 80% for seed 0), SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for the receiver reading the text. Cross-family transfer is as strong as within-family on the binding test and within 7 F1 of it on passages, and the linear map alone does nothing across families. The recipe fails for K2-Horizon-0.9B, a same-family sender with a different vocabulary and head dimension 64: the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1 on SQuAD. K2-0.9B differs from the receiver in vocabulary, layer count and head dimension at once; which difference matters is untested.

**Different contexts.** Table 5: HotpotQA bridge questions, sender and receiver each holding one of the two supporting paragraphs (the LCF-X setting of Rossi et al., 2026). The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9 / 39.7 / 42.1 over the three training seeds (mean 39.9, sd 2.1)—63% of the gap—while the deranged cache gives 29.9–32.1 (no help), the ridge map 26.3 and a zero cache 20.9. The channel carries the *other* paragraph, and only the trained map does.

Table 5. Different contexts: HotpotQA bridge questions,  $n = 400$ . Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1; the mapped cache is mean  $\pm$  sd over three training seeds (37.9 / 39.7 / 42.1), the deranged cache their range.

| receiver additionally gets                   | F1                               |
|----------------------------------------------|----------------------------------|
| nothing (own paragraph + question)           | 30.1                             |
| the sender’s paragraph as text               | 45.8                             |
| the sender’s paragraph as a mapped cache     | <b>39.9 <math>\pm</math> 2.1</b> |
| a deranged mapped cache                      | 29.9–32.1                        |
| the sender’s paragraph through the ridge map | 26.3                             |
| a zero cache                                 | 20.9                             |

Table 6. What the message costs. Message size is bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant,  $n = 300$ , seed-2 projector (full cache 75.3%). Rows with a smaller projector are separate 1,000-step training runs. The full sweep is in Appendix D.

| message                               | KB / token | Noma | SQuAD F1 |
|---------------------------------------|------------|------|----------|
| full mapped cache, bf16               | 144        | 75.3 | 53.0     |
| int8 fake-quantised                   | 72         | 76.0 | 52.1     |
| int4 fake-quantised                   | 36         | 75.0 | 51.6     |
| low-rank, rank 32                     | 78.7       | 75.0 | 45.3     |
| low-rank, rank 16                     | 39.3       | 75.0 | 29.6     |
| layers 18–35 only, zeros elsewhere    | 72         | 24.3 | 8.3      |
| 4 of 8 heads, zeros elsewhere         | 72         | 25.7 | 8.2      |
| layers 18–35, mean-vector fill        | 72.7       | 33.7 | 21.4     |
| layers 12–35, mean-vector fill        | 96.4       | 35.0 | 28.8     |
| projector bottleneck 32 (70 M params) | 144        | 64.3 | 53.7     |
| projector bottleneck 16 (62 M params) | 144        | 61.3 | 51.6     |

**What the message costs.** The full mapped cache is 36 layers  $\times$  8 heads  $\times$  2  $\times$  128  $\times$  bf16 = 144 KB per token, 75 MB for a 512-token prefix, against  $\sim$ 1 KB of text; the projector has 151 M parameters. Table 6 and Figure 4: two things shrink it for free and two do not. int8 and int4 fake-quantisation of the mapped cache keep binding accuracy (76.0 / 75.0 vs 75.3%) and SQuAD F1 (52.1 / 51.6 vs 53.0)—bytes  $\div$  4. A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. Dropping layers or heads with zeros in their place collapses transfer (layers 18–35 only: 24.3%; 4 heads: 25.7%), unlike the layer-band experiment where the other layers carried a *plausible* cache; an all-zero layer makes the receiver attend uniformly to nothing and drowns the question. Filling untransmitted layers with a one-vector summary (the position-mean of that layer’s mapped cache) recovers only part of it (layers 18–35: 33.7%; layers 12–35: 35.0%; SQuAD F1 21–29), so with this projector the message cannot be cut by layers: every receiver layer wants its own cache. Low-rank truncation of each (positions  $\times$  head-dim) matrix keeps the binding test (rank 32 / 16: 75.0%) but not passages (F1 45.3 / 29.6). Latency on one A100: the receiver re-reading 512 tokens takes 48 ms, the projector 24 ms (2.0 $\times$ ; the sender’s own read, 45 ms, is al-

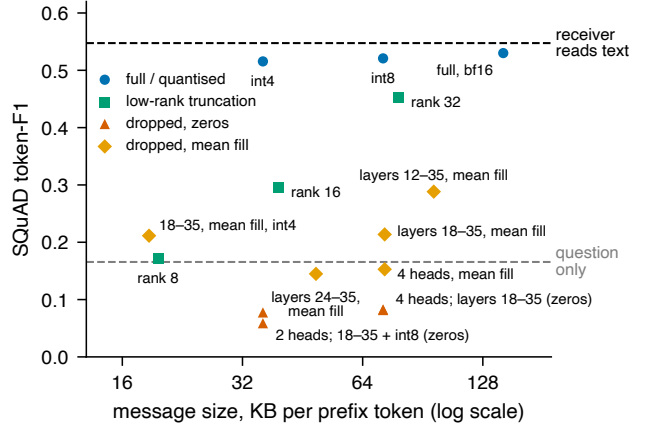


Figure 4. Message size against SQuAD F1 for every compression of the mapped cache we tried ( $n = 300$ ). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

Table 7. Handoff policy on 600 SQuAD items. Top: AUROC of each trigger at predicting the sender’s own failures (F1 < 0.5). Bottom: F1 of each arm when every item is handed off (rate 1).

| sender (3.7B) alone: F1 52.2, wrong on 35.5% of items |              |
|-------------------------------------------------------|--------------|
| trigger                                               | AUROC        |
| probe on the sender’s hidden state                    | <b>.749</b>  |
| semantic entropy (10 samples)                         | .689         |
| sample disagreement                                   | .683         |
| max-probability                                       | .582         |
| entropy                                               | .584         |
| receiver (7B) sees                                    |              |
|                                                       | F1 at rate 1 |
| the passage as text                                   | 55.7         |
| the mapped cache                                      | 51.8         |
| the sender’s answer + confidence word                 | 47.0         |

ready paid). But at 72 MiB per 512-token message a 1 Gbps link needs  $\sim$ 604 ms (int4:  $\sim$ 151 ms) against 24 ms of receiver compute saved, so the compute advantage does not carry over to a network handoff at this message size; and the quantisation result is fake-quantisation, not a packed transport. The message is therefore cheap in time and expensive in bytes; the honest framing is that the channel is worth it when the sender’s state is the point—which Section 4 tests.

### 3.1. When Should the Small Model Hand Off?

On 600 SQuAD items the 3.7B alone reaches F1 52.2 and gets 35.5% of items wrong (F1 < 0.5). A logistic probe on its pre-action hidden state (last prompt token, layers 12/18/24, nested 5-fold) predicts those failures with AUROC .749, better than its own semantic entropy (.689), sample disagreement (.683), max-probability (.582) or entropy (.584) (Table 7). But the 7B reading the passage reaches

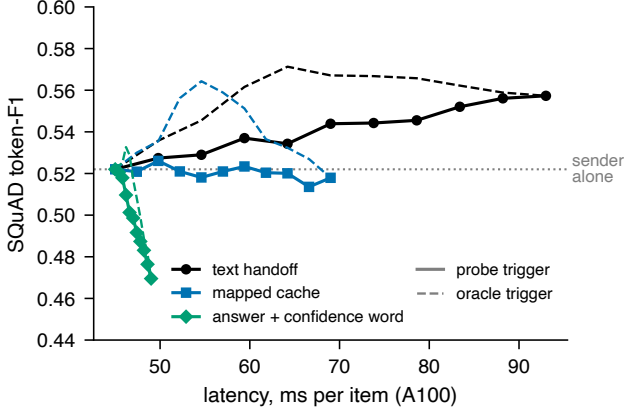


Figure 5. F1 against latency per item as the handoff rate goes from 0 (sender alone, 45 ms) to 1, for the three handoff arms under the probe trigger (solid) and an oracle trigger that hands off exactly the sender’s failures (dashed). 600 SQuAD items. Only the oracle gains; the probe-triggered text handoff climbs to the receiver’s own 55.7, the cache handoff stays flat.

only 55.7, the mapped cache 51.8 and the verbal handoff (the sender’s greedy answer plus a confidence word, defined in Section 4) 47.0, so no trigger makes a handoff pay at equal latency (Figure 5): the probe-triggered text handoff climbs from 52.2 to 55.7 as the handoff rate goes to 1, the cache handoff stays flat at 52, and only an oracle trigger gains (56.4 at rate .4). On this task the receiver is not enough better than the sender for a handoff of any kind to be worth it; the probe result stands, the policy result waits for a task with a real gap.

#### 4. Does the Cache Carry How Sure the Sender Was?

**Why a text handoff launders uncertainty, and what a state handoff can keep.** Sender  $S$  reads context  $c$  and holds a belief  $p_S(y | c)$  over answers; it hands off through a message  $m = f(c, S)$ , and the receiver produces  $p_R(y | m, q)$ . With  $m = \arg \max_y p_S(y | c)$ ,  $m$  takes at most  $|Y|$  values, so the information  $m$  carries about  $p_S$  is bounded by  $\log |Y|$  nats however peaked or flat  $p_S$  is: two senders with the same mode and entropies 0.1 and 2.0 send the same message. A verbalised confidence word adds at most  $\log k$  nats ( $k = 3$  levels here) and relies on the sender’s verbal calibration, which is poor for base models (Tian et al., 2023; Yona et al., 2024). This is the mechanism behind confidence laundering (Shi et al., 2026). With  $m$  the KV cache of  $S$  at every layer and position,  $m$  is a sufficient statistic for  $S$ ’s next-token distribution, so  $I(m; p_S)$  is not bounded by  $\log |Y|$ . What the receiver recovers is limited by two losses: *map loss*—the map  $g$  is fit on continuation likelihood, and a learned map trained on point targets contracts predictive variance (Cui & Pei, 2026); and *read loss*— $R$  reads the mapped cache with its own heads, from its later layers (Table 2). The bound limits information about the sender’s *distribution* (which alternatives, how spread), not a detector’s AUROC: a thresholded label could in principle give a perfect

reader AUROC 1 against a thresholded target. The measured gap between the verbal and cache arms is therefore about how the receiver *uses* a label versus a state, not about channel capacity alone. Predictions: (1) the cache handoff can approach the text-reading receiver’s AUROC, its shortfall being map loss; (2) neither can exceed the ceiling set by how far  $R$ ’s first-token entropy correlates with  $S$ ’s semantic entropy at all, which is why absolute AUROCs are  $\sim 0.6$  and the comparison must be relative.

**Protocol.** Following the knowledge-conflict and semantic-entropy protocols (Longpre et al., 2021; Xie et al., 2024; Farquhar et al., 2024), each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold sentence deleted). We keep the 377 items the receiver answers correctly from clean text. The sender’s uncertainty is its semantic entropy over 10 sampled answers (mean 1.10 / 1.23 / 1.96 across the three variants, so the sender does notice). The receiver’s uncertainty is its entropy at the first answer token. The text handoff with a confidence word (*verbal*) is the text channel doing its best to carry uncertainty: the sender writes its greedy answer and “confidence: high / medium / low”, and the receiver reads only that. The word is derived from the sender’s *actual* semantic-entropy terciles (an oracle label, favourable to the text channel) and the numeric variant from its sample agreement; the sender does not verbalise its own confidence. The drop ratio is the cache path’s decrease in  $P(\text{gold})$  from clean to removed divided by the text path’s decrease:  $(.279 - .130) / (.437 - .162) = .54$ . It is a relative drop in confidence, not a fraction of uncertainty information retained, and the two paths start from different confidence levels.

**Result.** Table 8 and Figure 6. The mapped cache beats the verbal handoff in the same direction on every measure: it tracks the sender’s uncertainty better (AUROC  $+.03$  to  $+.06$  on the three variants), its drop ratio is .54 where the verbal handoff’s is .30, and it answers better on clean and contradicted passages (F1 59.8 / 49.0 vs 55.0 / 37.7). Writing the sender’s confidence as a number instead of a word (its sample agreement, e.g. “0.8”) does not help the text channel (AUROC .548 / .532 / .563, drop ratio .32), and neither does the strongest text message we can build from the sender’s samples—every answer cluster with its frequency, e.g. “1929 (0.6); 1932 (0.3); 1935 (0.1)” (AUROC .495 / .489 / .626, drop ratio .44, F1 55.4 / 40.7 / 22.3): the receiver answers about as well from it as from the confidence word and its own confidence tracks the sender no better, except on removed passages where it matches the cache. With paired bootstrap intervals over passages (344 passages, 2,000 resamples), the AUROC gap cache–verbal is significant on contradicted passages ( $[-.008, .117]$ ) and not on clean ( $[-.027, .094]$ ) or

Table 8. Sender uncertainty through each handoff; 377 SQuAD items the receiver answers correctly from clean text. AUROC is of the receiver's first-token entropy against the sender's semantic entropy (thresholded at its median per variant); the drop ratio is the receiver's  $P(\text{gold})$  loss from clean to removed relative to the text path's.

| receiver sees                                                     | $P(\text{gold})$          | entropy         | AUROC vs sender SE        | drop ratio |
|-------------------------------------------------------------------|---------------------------|-----------------|---------------------------|------------|
|                                                                   | clean / contra. / removed | clean / removed | clean / contra. / removed |            |
| text                                                              | .437 / .290 / .162        | 1.14 / 3.03     | .597 / .615 / .671        | 1 (def.)   |
| text + answer + confidence word (verbal)                          | .253 / .193 / .171        | 2.88 / 2.80     | .552 / .542 / .592        | .30        |
| text + answer + numeric confidence (sample agreement, e.g. "0.8") | .258 / .197 / .170        | 2.93 / 2.93     | .548 / .532 / .563        | .32        |
| text + all candidate answers with their sampled probabilities     | .264 / .210 / .144        | 2.68 / 2.63     | .495 / .489 / .626        | .44        |
| <b>mapped cache</b>                                               | .279 / .249 / .130        | 1.78 / 3.16     | <b>.585 / .605 / .624</b> | <b>.54</b> |
| deranged cache                                                    | .105 / .106 / .105        | 3.88 / 3.88     | .536 / .488 / .564        | 0          |
| zero cache                                                        | .034 / .033 / .037        | 6.41 / 6.18     | .496 / .481 / .502        | 0          |
| moment-matched random cache                                       | .005 / .004 / .005        | 7.24 / 7.30     | .457 / .472 / .580        | 0          |
| question only                                                     | .127                      | 3.69            | .532 / .479 / .603        | 0          |
| mapped cache, KL-trained projector                                | .272 / .241 / .118        | 2.38 / 3.76     | .590 / .589 / .611        | .56        |
| mapped cache, sender-entropy-matched projector                    | .247 / .222 / .107        | 2.23 / 3.64     | .596 / .587 / .627        | .51        |

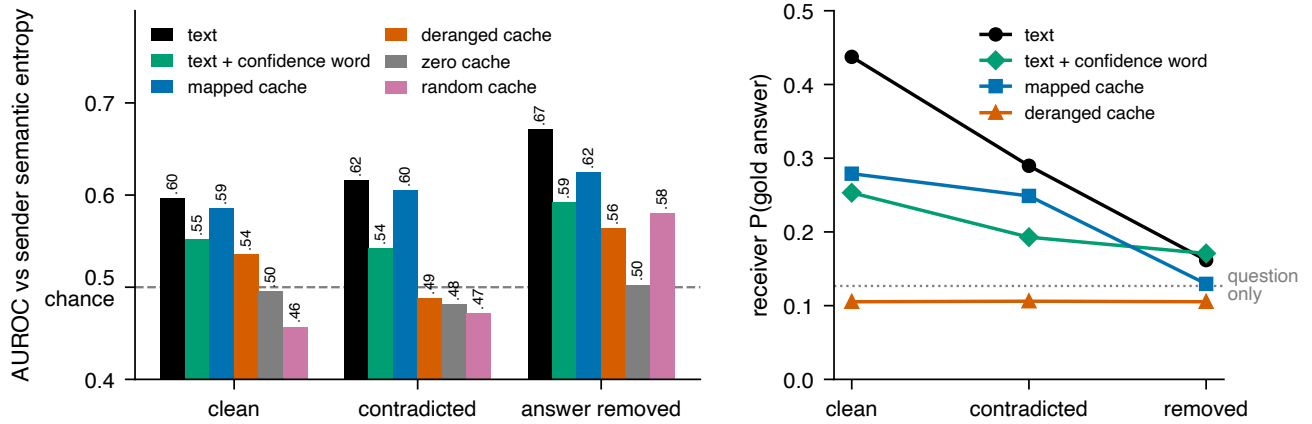


Figure 6. Left: AUROC of the receiver's first-token entropy against the sender's semantic entropy, per handoff and passage variant (377 items). The content-free controls sit at .46–.54 on clean and contradicted passages and at .56–.60 on removed ones; the mapped cache sits just below the receiver reading the text and above the text handoff with a confidence word. Right: the receiver's  $P(\text{gold})$  from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.

removed ( $[-.024, .086]$ ); cache–question-only is significant on contradicted ( $[-.065, .180]$ ) and borderline on clean ( $[-.005, .111]$ ); on removed passages the text path is significantly above the cache ( $[-.091, -.002]$ ). The direction is consistent; the size is modest and established on one of three variants.

**The label carries the signal; the receiver does not read it.** The confidence word is built from the sender's own semantic-entropy terciles, so as a score it predicts the binary target almost perfectly without any receiver: AUROC .955 / .942 / 1.000 on the three variants (the numeric label .870 / .848 / .735). Yet the receiver's first-token entropy after reading that word reaches only .552 / .542 / .592. The text channel is not capacity-limited here; the receiver does not turn a label into its own spread. The cache arm's higher AUROC therefore says that a state changes the receiver's confidence in a way a label does not, not that the state carries more information than the label. This is the comparison the paper can support, and it is narrower than "text cannot carry

uncertainty".

**Within-item intervention.** Cross-item correlations admit a common cause: a hard question makes both models uncertain whether or not anything crosses. The variants allow a paired test with the question held fixed and only the sender's passage changed (clean  $\rightarrow$  removed). Per item, the receiver's entropy change tracks the sender's semantic-entropy change through the mapped cache (Spearman .280; same sign on 88% of items) about as well as when the receiver reads the text (.322; 90%), weakly through the confidence word (.146; 49%), and not at all through a cache from another item (.004; 47%), which shares the question and rules out the common cause. Under a contradiction the effect is smaller (.208 vs .325 for text). The receiver's confidence moves with what the sender saw, and it moves because of the cache's content.

Three readings. (i) The receiver's confidence through the cache tracks the *sender's* semantic entropy about as well as

it does when the receiver reads the text itself (AUROC .585 vs .597 on clean), and the absolute level is bounded by the text path, not by the channel. On removed passages question-only gives .603, the moment-matched random cache .580 and the deranged cache .564 against the mapped cache’s .624 (on clean .532 / .457 / .536; on contradicted .479 / .472 / .488). The cache’s increment over question-only is +.053 on clean, +.126 on contradicted and +.021 on removed passages: on removed passages the question alone already carries most of the signal. (ii) When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise: uncertainty crosses, attenuated. Whether the attenuation comes from the map, from how the receiver reads a foreign cache, or from the mismatch between the training objective and the measured quantity is not isolated here; distillation work (Cui & Pei, 2026) reports a similar contraction of predictive variance in its setting, but that is an observation to compare against, not a theorem that covers this map. The ten-sample semantic entropy is also coarse (28–40 distinct values per variant; maximum  $\ln 10$ ), which the intervals above absorb but do not remove. (iii) Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ( $P(\text{counter})$  .32 >  $P(\text{gold})$  .29), while the cache path keeps the original ahead (.25 > .22) and its F1 is higher (49.0 vs 45.2). Across all conditions the cache path is less confident than the text path on clean passages ( $P(\text{gold})$  .28 vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache; per-channel calibration is the obvious fix.

**Two negative results on training the loss away.** A first attempt did not work: adding  $\text{KL}(\text{text-path} \parallel \text{cache-path})$  on the receiver’s next-token distributions to the projector’s objective (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1 (Table 8). Matching the receiver’s own text-path distribution is the wrong target: what is lost is the *sender’s* spread, which the receiver’s text path does not contain either. A sender-side target—matching the receiver’s per-token entropy through the cache to the sender’s own per-token entropy on the training text—was the second attempt (weight 1, 1,000 steps): binding accuracy holds (72.3%; Appendix A), SQuAD F1 falls to 45.3, and the uncertainty measures do not move (AUROC .596 / .587 / .627; drop ratio .51). Neither receiver-side objective recovers the sender’s spread; what is compressed is compressed by the map’s capacity or by how the receiver reads a foreign cache, and separating those two is the next paper’s first experiment.

**Belief structure and where the loss occurs.** Two diagnostics address the objection that sender semantic entropy and receiver first-token entropy are different

Table 9. Continuation retention is the wrong headline. Retention =  $(\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$  on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

| projector                 | retention | retention, sender |
|---------------------------|-----------|-------------------|
|                           |           | text deranged     |
| ridge top-3               | −0.30     | −0.66             |
| + residual                | 1.23      | 0.31              |
| residual, deranged sender | 0.77      | 0.72              |

measurements. The first is the Spearman correlation, per arm, between the sender’s and the receiver’s distributions over the sender’s sampled answer candidates. The second fits a probe for the sender’s semantic entropy on the raw sender cache, the mapped cache, and the receiver’s hidden state after reading it, to see whether the spread is lost in the map or in the read. [TODO: numbers from `analysis/cloud/night_a/diag_unc_*.json` and `confidence_belief.json` when they arrive.]

## 5. Controls

Three controls bound what the receiver actually gets from the cache, and a fourth shows why the usual headline number cannot.

**Deranged cache.** The *derange* arm of Table 1: a mapped cache from another episode with the same layout but a different value for the queried name. Accuracy falls to 4.6% and the receiver answers with the other episode’s value 63.7% of the time. The receiver is not ignoring the cache; it is reading it.

**Deranged-sender projector.** A projector trained with the identical recipe but with the sender reading text unrelated to the receiver’s continuation. Its residual cannot learn a content channel, only a soft prompt. It reaches 23.4% on the binding test (Table 1, row 3) and 15.1 F1 on SQuAD (Table 3). It is not content-free—its ridge base is weakly content-dependent (ridge alone 21.7%)—so it controls for the *learned* part only.

**Layer-band injection.** Table 2: injecting this episode’s cache into layers 0–17 alone gives chance (10.4%); layers 18–35 alone give 51.8%; no six-layer band suffices.

**Continuation retention is the wrong headline.** Following the closed-form-map literature (Heo et al., 2026) we also report retention (Table 9, Figure 7). Theirs is an accuracy ratio (transferred accuracy over the receiver’s own); ours is the fraction of the continuation-loss gap recovered, so the table critiques our own metric, not theirs. A projector trained with the sender reading the *wrong* text—a learned soft prompt on a weakly content-dependent ridge base—recovers 77% of the oracle gap, and almost all of it (0.72)

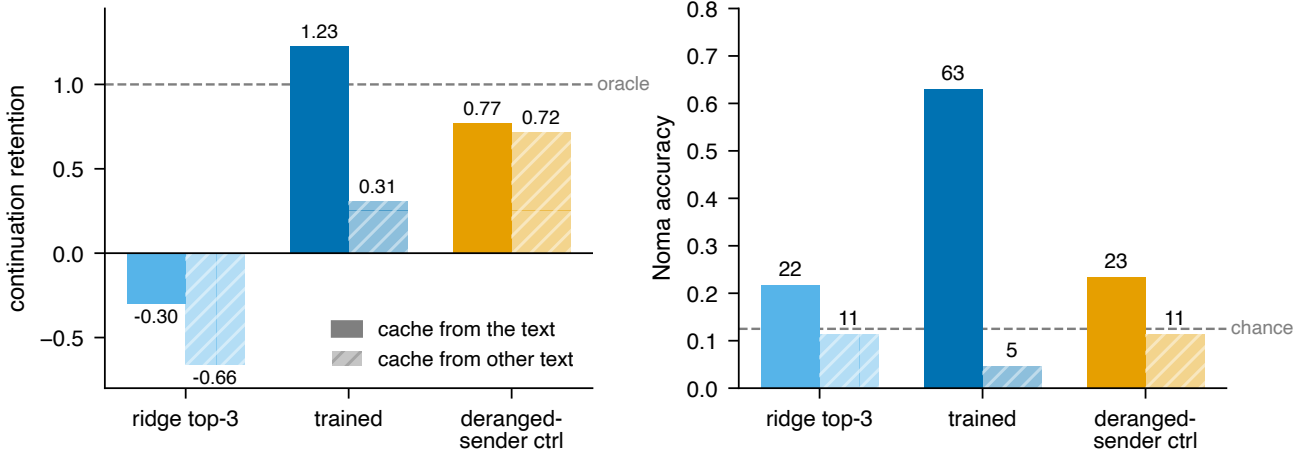


Figure 7. Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ( $n = 1000$ , 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the deranged-sender projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

survives when its input is deranged at test time. The trained projector’s 1.23 decomposes into  $\sim 0.3$  of the same prompt-like effect and  $\sim 0.9$  that disappears when the sender’s text is swapped. Retention above 1 is therefore not evidence of communication, and retention alone cannot distinguish a channel from a prompt; the binding test with its derange arm can.

## 6. Related Work

Cache-to-cache communication (C2C; Fu et al., 2025) and Latent Cache Flow (Rossi et al., 2026) train fusers between models of similar size; a closed-form KV map (Heo et al., 2026) reports 44–98% continuation retention across sizes; a causal audit (Cheng et al., 2026) finds that deranged caches lose only 0.4 points in such setups, which is the objection our derange and deranged-sender arms answer. A second audit (Zhang & Emu, 2026) decomposes the aggregate effect of a latent channel into “sender content” and “any message present” terms with boundary-replacement controls; our deranged-sender projector isolates the second term directly. Dual-Cache (Liu et al., 2026a), CacheBridge (Qu et al., 2026) and HyLaT (Mou et al., 2026) extend cache transfer to heterogeneous pairs and hybrid latent–text protocols. CIPHER (Pham et al., 2023) had agents debate through embeddings rather than tokens; Interlat (Du et al., 2025) has agents communicate entirely in latent space; and K-V cache alignment (Dery et al., 2026) learns a map between two models’ caches, the closest setup to ours, without a content-free control or an uncertainty test. In the last six weeks, a universal context-reuse layer (Li et al., 2026) makes one model’s KV state consumable by another across scale, tokenizer and family (including Llama-3.1-70B  $\rightarrow$  Qwen2.5-7B) but reports no content-free control or uncertainty test; StateBridge

(Peng et al., 2026) aligns final-layer hidden states into the receiver’s input-embedding space with a closed-form orthogonal map and no training, a lighter hidden-state-prefix interface that we do not compare against; LatentPort (Villani, 2026) hands attention KV plus recurrent state from a 4B to a 9B hybrid model without prefix replay, scored on continuation NLL rather than question answering; MentorPulse (Liu et al., 2026b) finds that a static one-shot latent handoff can fall below no guidance on long-form generation, a scope limit our short-answer setting does not probe; and a preregistered alignment ladder between trained language-model cells (Marincat, 2026) runs a leakage-controlled causal audit close in spirit to our derange arms. The Replay Gap (Gonuguntla, 2026) and Confidence Laundering (Shi et al., 2026) argue that handoffs must be evaluated on branched rollouts and must carry uncertainty; the second is the question Section 4 answers for one channel, the first remains a next step. On the uncertainty side we borrow the knowledge-conflict construction of Longpre et al. (2021) and Xie et al. (2024), semantic entropy as the sender’s uncertainty (Farquhar et al., 2024), the finding that verbalised confidence is poorly calibrated (Tian et al., 2023; Yona et al., 2024), and the result that distillation-style maps contract predictive variance (Cui & Pei, 2026). These works are concurrent with ours; we treat them as the surrounding conversation, not as prior results this paper improves on.

## 7. Limitations

The results show that a specific fact can cross from a 3.7B cache into a 7B without text, that the crossing depends on the cache’s content, that the receiver reads it out from its later layers, and that the mapped cache keeps more of the sender’s uncertainty than a text handoff with a confidence word. They

do not show that the channel is cheaper than text in bytes (144 KB per token against  $\sim 1$  KB of text for a passage; quantisation buys  $4\times$  and nothing else we tried is free), that it carries anything richer than a single fact or a short answer span, or that it survives long-form generation, where Liu et al. (2026b) report that a static latent handoff can hurt. The uncertainty result rests on 377 items of one dataset, one receiver, one sender-uncertainty measure and one design of confidence word; the absolute AUROCs are  $\sim 0.6$  for every channel including text, so the claim is relative. The recipe crosses one family boundary (Qwen3-4B) and fails on K2-Horizon-0.9B, for a reason not yet isolated (Section 3). The setting is context reuse—the sender’s cache of a passage prefix—not a mid-task handoff of an agent’s reasoning or tool state. Two receiver-side objectives for training the confidence loss away failed (a KL to the receiver’s text path and a sender-entropy match), and whether the loss is in the map or in the read is untested. The handoff-policy result is negative on SQuAD for a reason specific to the task: the receiver is only 3.5 F1 better than the sender, so no trigger, not even the .749-AUROC probe, makes a handoff pay; a task with a real sender–receiver gap is needed before the policy question can be answered.

## 8. Conclusion

A text handoff is a point estimate and cannot carry more than  $\log |Y|$  nats about how sure the sender was; a confidence word adds  $\log 3$ . A learned map from a small model’s KV cache to a larger model’s carries the content (63–73% binding transfer, 53.0 F1 on SQuAD, 71.7% across families) and, on top of it, more of the sender’s uncertainty than the text handoff with a confidence word does (AUROC .585–.624 vs .542–.592; drop ratio .54 vs .30). The controls matter more than the headline: the wrong cache is followed, the deranged-sender projector fails the test, retention cannot tell a channel from a prompt, and the signal is read out late in the receiver. What is lost is the sender’s spread, and two receiver-side training objectives—matching the receiver’s own text-path distribution and matching the sender’s entropy—do not recover it; whether the loss sits in the map or in the read is the next experiment.

## Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here. One consideration specific to cache handoff: a cache that a receiver “believes” (Section 5) is also a cache that can be tampered with; deployments that move caches between devices should authenticate them as they would any other input.

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## A. Ablations on the 6-Name Variant

All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, 6 names,  $n = 300$  episodes, chance 12.5%. Accuracy in percent. The two 1,000-step rows with an added uncertainty term (KL, sender-entropy matching) use the full 151 M-parameter projector.

Table 10. Projector ablations on the 6-name Noma variant.

| projector                                         | project | derange | follow |
|---------------------------------------------------|---------|---------|--------|
| ridge top-3 (closed form)                         | 22.0    | 10.7    | 30.3   |
| + residual, 300 steps                             | 56.3    | 6.3     | 53.7   |
| + residual, 1500 steps, seed 0                    | 71.3    | 4.3     | 71.7   |
| + residual, 1500 steps, seed 1                    | 73.0    | 3.0     | 74.0   |
| + residual, 1500 steps, seed 2                    | 75.3    | 2.3     | 76.0   |
| + residual, 1500 steps, ridge frozen              | 59.0    | 7.0     | 59.3   |
| + residual, 1500 steps, aligned-layer ridge       | 32.7    | 8.0     | 34.7   |
| + residual, 1000 steps, bottleneck 32             | 64.3    | 5.7     | 62.3   |
| + residual, 1000 steps, bottleneck 16             | 61.3    | 7.7     | 60.0   |
| + residual, 1000 steps, + KL(text    cache)       | 66.0    | 5.3     | 63.3   |
| + residual, 1000 steps, + sender-entropy matching | 72.3    | 3.0     | 73.3   |

## B. Other Attributes and Reversed Roles

Table 11. Binding transfer with city and animal attributes ( $n = 500$ , 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names,  $n = 300$ ).

| setting  | projector            | none | project | derange | follow |
|----------|----------------------|------|---------|---------|--------|
| city     | ridge top-3          | 14.2 | 19.8    | 11.8    | 18.4   |
| city     | trained              | 14.2 | 61.8    | 6.4     | 60.2   |
| city     | deranged-sender ctrl | 14.2 | 15.4    | 11.4    | 18.6   |
| animal   | ridge top-3          | 12.2 | 12.8    | 12.0    | 11.2   |
| animal   | trained              | 12.2 | 67.2    | 5.2     | 64.6   |
| animal   | deranged-sender ctrl | 12.2 | 12.4    | 12.8    | 9.8    |
| reversed | ridge top-3          | 11.7 | 29.0    | 12.3    | 31.3   |
| reversed | trained              | 11.7 | 63.3    | 5.7     | 63.0   |

## C. Distance Sweep and Layer Bands

Table 12. Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant;  $n = 1000$  at pad 0,  $n = 500$  otherwise.

| projector            | 0    | 64   | 128  | 256  | 512  |
|----------------------|------|------|------|------|------|
| ridge top-3          | 21.7 | 22.2 | 22.4 | 22.8 | 24.2 |
| trained              | 63.0 | 61.0 | 57.0 | 62.4 | 68.8 |
| deranged-sender ctrl | 23.4 | 25.0 | 19.8 | 18.2 | 13.8 |

## D. Compression Sweep

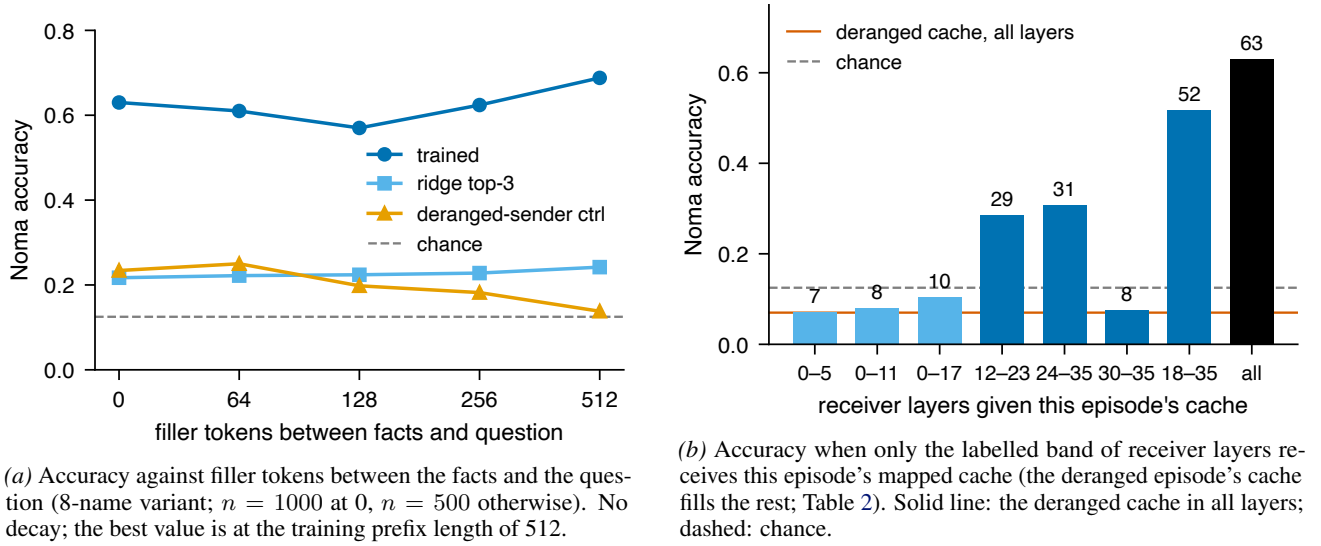


Figure 8. Distance sweep and layer-band injection.

Table 13. Every compression of the seed-2 projector's mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant,  $n = 300$ , percent; SQuAD:  $n = 300$ , token-F1.

| spec                          | KB / token | Noma project | Noma derange | SQuAD F1 |
|-------------------------------|------------|--------------|--------------|----------|
| none (full, bf16)             | 144        | 75.3         | 2.3          | 53.0     |
| int8                          | 72         | 76.0         | 2.7          | 52.1     |
| int4                          | 36         | 75.0         | 3.7          | 51.6     |
| rank 32                       | 78.7       | 75.0         | 2.3          | 45.3     |
| rank 16                       | 39.3       | 75.0         | 2.3          | 29.6     |
| rank 8                        | 19.7       | 51.7         | 7.0          | 17.2     |
| layers 18-35, zeros           | 72         | 24.3         | 9.7          | 8.3      |
| layers 18-35, zeros, int8     | 36         | 24.7         | 9.7          | 7.8      |
| 4 heads, zeros                | 72         | 25.7         | 8.7          | 8.2      |
| 2 heads, zeros                | 36         | 22.3         | 9.7          | 5.9      |
| layers 12-35, mean fill       | 96.4       | 35.0         | 11.0         | 28.8     |
| layers 18-35, mean fill       | 72.7       | 33.7         | 11.7         | 21.4     |
| layers 18-35, mean fill, int4 | 18.7       | 33.3         | 11.7         | 21.1     |
| layers 24-35, mean fill       | 48.9       | 29.3         | 12.0         | 14.5     |
| 4 heads, mean fill            | 72.7       | 23.0         | 11.3         | 15.3     |